Master table

ci-tec | README | 12.10.2026 | Master schema v7

Use **Bio_O_Ton_Master.csv**, one row per recording. No separate ci-tec table is required. This is the latest recorded inventory, not a guarantee that processing is finished. Original metadata source: dawn-chorus-soundscape.csv.

Column	Meaning
dawn_chorus_id	Recording ID.
lon, lat	WGS84 coordinates in decimal degrees.
grid_100m_id, grid_10m_id	Assigned grid cells. A grid ID does not imply that LRT formation data are available.
datetime_local	German local date and clock time in Europe/Berlin, stored without a timezone suffix. Its origin is recorded separately in timestamp_source.
datetime_utc	UTC instant of the selected recording time. When localtimes is absent, this is the datetime instant from the original Dawn Chorus CSV.
timestamp_source	Exact values: localtimes for an explicitly supplied local clock; datetime for the UTC fallback. A populated datetime_local does not imply source localtimes.
time_has_issues	False only for source localtimes; True for datetime, absent or unknown provenance. Invalid local times are separately rejected by metadata checks.
duration_seconds	Checked audio duration in seconds; null if unavailable.
inside_lrt_polygon	Recording point lies within an included LRT polygon.
grid_100m_has_majority_formation, grid_10m_has_majority_formation	Majority formation is available for this cell; different from a direct point/LRT intersection.

Grid embeddings, the LRT GeoPackage and grid reference data are separate products. Recording/bioacoustic flags do not certify their completeness.

Recording time: source and German clock

The original Dawn Chorus column is named **localtimes** (the supplied local time). **Only when this field is absent, empty or whitespace-only** is the original **datetime** value used as a UTC instant and converted to **Europe/Berlin**. German winter time is UTC+1 and summer time is UTC+2; the conversion uses the timezone rules for the recording date, including date changes at midnight. Do not add a fixed offset or convert the resulting German clock a second time.

Examples without localtimes: 2025-01-15T06:00:00Z becomes 2025-01-15 07:00:00; 2025-07-15T06:00:00Z becomes 2025-07-15 08:00:00. A UTC evening can become the following German calendar day.

When localtimes is supplied, its local clock retains priority and is interpreted as German time. These recordings are unchanged by the fallback rule. A nonempty but invalid or unresolved localtimes value is flagged; it is not silently replaced with datetime. Source conflicts remain visible in the Step-1 time log.

Audit on 28.09.2026: all 3,288 recordings without localtimes already match this UTC-to-Germany conversion in both published master copies, including UTC, German date and clock, and the source label. No timestamp correction was required. Of these, 3,264 use UTC+2 and 24 use UTC+1. The published master has 108,578 recordings with source localtimes and 3,288 with source datetime.

Clean-data filters from the 9 October PDF

Apply the filters needed by your analysis:

```
time_has_issues=False
sound_exists=True
duration_seconds >= 40
sound_has_issues=False
weather_point_exists=True
weather_point_has_issues=False
sentinel_exists=True
sentinel_has_issues=False
```

The 40-second threshold is an analysis choice and can be replaced. Domain flags

do not automatically invalidate the entire recording. For training in cells with

a majority formation add grid_100m_has_majority_formation=True; for direct

point-in-LRT training add inside_lrt_polygon=True. Grid IDs alone certify neither.

Recording-time confidence and weather use

Use weather data only for recordings with time_has_issues=False, equivalently timestamp_source == "localtimes". UTC fallback timestamps are converted correctly but can differ from the actual supplied recording clock by up to two hours according to the PDF; conversion alone does not establish confidence in the source.

This source restriction supplements weather availability and quality checks. In schema v7, combined readiness flags include it. It does not automatically exclude the recording from audio-only work, and the audio-only bioacoustic readiness flag remains independent of time confidence. Recordings and stored files remain in the shared master.

Availability and issues

_exists: True means the corresponding files were recorded as present on LSDF. These flags describe the recorded inventory, not a live file check.

_has_issues: True means a problem or missing required data/check results. It can be True whether _exists is True or False.

Empty means no check result is available.

_issue_codes gives the reason. Multiple codes are separated by a pipe; details may follow a colon. **_has_issues is not a severity level or a universal exclusion rule.** Rules below concern the affected file/domain, not automatically the entire recording. Unknown codes or a True issue flag without codes require review.

Columns	Data
sound_exists, sound_has_issues	Audio file and audio checks.
sentinel_exists, sentinel_has_issues	Sentinel-2 files and technical checks.
weather_point_exists, weather_point_has_issues	Weather file for the recording and its checks.

Sound examples:

- True / False: file found, no issues reported.

- True / True: file found, but a technical property or decoding check failed.

- False / True: file missing.

Sound codes and suggested use

sound_issue_codes	Meaning / consequence
duration_not_60s	Outside 59-61 seconds. **Information**; decide in application-specific preprocessing. Not a general exclusion.
missing_file	File absent: unavailable.
empty_file, no_audio_stream_found, no_audio_frames_decoded	No usable audio in the checked file: exclude that file pending repair.
pyav_decode_failed, invalid_or_missing_sample_rate, invalid_or_missing_channel_count, invalid_duration, duration_unavailable	Technical check failed or a property could not be established: hold for review. Does not prove the original source is permanently corrupt.
multiple_audio_streams	Select the intended stream before use.
filename_does_not_match_<id>_audio_pattern	File-to-ID association needs clarification.
source_stat_failed, copy_failed, WinError	Access/copy problem; investigate connection and download before concluding source corruption.

PDF audio filter: sound_exists=True, duration_seconds >= 40, sound_has_issues=False. A pure duration_not_60s note does not set the issue flag. Shorter or longer files can be usable; change the duration threshold for your analysis. Technical codes still require review.

Point weather codes and suggested use

Expected window: ten preceding complete German calendar days plus the recording day, hourly. Normally 264 rows; across spring DST 263, across autumn DST 265. The repeated local autumn hour is valid; a timezone-free timestamp alone cannot distinguish its two UTC instants.

weather_point_issue_codes	Meaning / consequence
missing_file	No weather file. Cause is not established by this code.
missing_value	At least one required value is missing. **Information only:** this code alone does not set weather_point_has_issues. Check variables/hours needed by the application; do not fill values blindly.
recording_date_changed_recheck_required	Recording date changed while previous results were retained. Temporal association is unconfirmed: hold until rechecked.
unexpected_time_interval	Full timestamp sequence and multiplicities differ from the expected window. May mean missing, extra or shifted hours. Correct DST transitions are accepted by the current inventory. **Check/correct before use.**
unexpected_time_window, unexpected_row_count	Wrong boundaries or row count: check/correct before use.
duplicate_datetime	More repeats than allowed by the expected window, not the legitimate repeated autumn hour. Check/correct before use.
missing_required_column, missing_datetime_column, unparseable_datetime, invalid_metadata_datetime	Structure or time basis is unreliable: hold until clarified.
empty_file, read_error	Empty or unreadable; an exception suffix may follow read_error. File currently unusable.

weather_point_issue_codes	Meaning / consequence
implausible_value	Temperature/humidity range check failed. Investigate values and source before deciding usability.

PDF weather filter: time_has_issues=False, weather_point_exists=True and weather_point_has_issues=False. This permits missing_value as an informational code; verify the variables/hours your analysis needs. A stricter application can additionally require no codes. Additional time/structure errors remain blocking.

Sentinel

Code / field	Meaning / consequence
Sentinel: missing_file, score_without_tif	Image missing; a score does not replace it.
Sentinel: too_cloudy_sentinel_not_available	Quality score is exactly 0: too cloudy, Sentinel data unavailable for use. The numeric score stays 0; sentinel_has_issues=True. sentinel_exists still reports physical file presence.
Sentinel: quality_score_missing	Image may exist, but its quality assessment is missing.
Sentinel: empty_file, raster_read_failed	Image empty or unreadable: exclude pending repair.
inventory_not_run, qc_not_run	Check results unavailable; not equivalent to passing QC.

Weather and Sentinel do not automatically exclude a recording from audio/LRT-only work. Photos are not a filter criterion for the stated ci-tec priorities.

Processing completion is separate from quality

is_final does not currently exist. ready_for_general_analysis is not a ci-tec release flag: it requires trusted time provenance plus unflagged sound, weather and Sentinel together. It does not enforce an analysis-specific audio duration threshold; apply duration_seconds >= 40 separately for the PDF clean filter.

Completion requires finished processing and checks for the required products, a specified input/code version and no outstanding repairs. Missing data and suitability remain separate. A permanent release applies only to a frozen version; later versions can reveal new errors. The shared master retains all rows.